

Master Table of Contents for Digital Electronics Notes

From Basic Gates to Complex Digital Circuits

0. Standard Note-Making Format for Every Circuit

For every digital circuit, include the following wherever applicable:

0.1 Basic Details

- Circuit name
- Purpose / function
- Type of circuit:
 - Combinational
 - Sequential
 - Synchronous
 - Asynchronous
 - Arithmetic
 - Data routing
 - Memory
 - Programmable logic
 - Input and output variables
 - Active-high / active-low convention

0.2 Representations to Include

- Logic symbol
- Gate-level diagram
- Block diagram
- Truth table
- Boolean expression
- Simplified Boolean expression
- Canonical SOP form
- Canonical POS form
- Minterm notation
- Maxterm notation
- K-map simplification
- Timing diagram
- Waveform behaviour
- HDL code, Verilog/SystemVerilog if required
- Applications
- Advantages and limitations

- Common exam mistakes

0.3 Sequential Circuit-Specific Details

- State table
 - State diagram
 - State equation
 - Characteristic table
 - Excitation table
 - Transition table
 - Present state and next state table
 - Timing diagram
 - Clock edge behaviour
 - Setup time
 - Hold time
 - Propagation delay
 - Race condition
 - Metastability possibility
 - Reset and preset behaviour
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1. Introduction to Digital Electronics

1.1 Signals

- What is a signal?
- Analog signal
- Digital signal
- Continuous-time signal
- Discrete-time signal
- Continuous-amplitude signal
- Discrete-amplitude signal
- Periodic and non-periodic signals
- Deterministic and random signals
- Noise in signals

1.2 Need for Digital Signals

- Noise immunity
- Storage capability
- Processing capability
- Reliability
- Reproducibility
- Programmability
- Ease of transmission
- Error detection and correction

1.3 Analog vs Digital Systems

- Analog system definition
- Digital system definition
- Comparison between analog and digital systems
- Advantages of digital systems
- Limitations of digital systems
- Mixed-signal systems
- Real-world examples

1.4 Basic Digital Concepts

- Bit
- Nibble
- Byte
- Word
- Logic 0 and Logic 1
- Voltage levels
- Logic high
- Logic low
- Active-high signals
- Active-low signals
- Positive logic
- Negative logic

1.5 Switch and Bit Intuition

- Mechanical switch as digital input
 - Open switch
 - Closed switch
 - Pull-up resistor
 - Pull-down resistor
 - Switch bouncing
 - Debouncing circuits
 - Digital abstraction from physical switches
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2. Number Systems

2.1 Introduction to Number Systems

- Radix / base
- Positional number system
- Weighted number system
- Non-weighted number system
- Integer part
- Fractional part
- Most significant bit
- Least significant bit

2.2 Decimal Number System

- Base-10 representation
- Decimal weights
- Decimal arithmetic

2.3 Binary Number System

- Base-2 representation
- Binary weights
- Binary place values
- Binary point
- Unsigned binary numbers

2.4 Octal Number System

- Base-8 representation
- Octal weights
- Relation between binary and octal

2.5 Hexadecimal Number System

- Base-16 representation
- Hexadecimal digits
- Relation between binary and hexadecimal

2.6 Number System Conversions

- Decimal to binary
- Decimal to octal
- Decimal to hexadecimal
- Binary to decimal
- Octal to decimal
- Hexadecimal to decimal
- Binary to octal
- Octal to binary
- Binary to hexadecimal
- Hexadecimal to binary
- Octal to hexadecimal
- Hexadecimal to octal
- Fractional number conversions
- Repeated division method
- Repeated multiplication method

2.7 Binary Arithmetic

- Binary addition
- Binary subtraction
- Binary multiplication
- Binary division

- Binary increment
- Binary decrement
- Overflow
- Carry
- Borrow

2.8 Octal Arithmetic

- Octal addition
- Octal subtraction
- Octal multiplication
- Octal carry and borrow rules

2.9 Hexadecimal Arithmetic

- Hexadecimal addition
 - Hexadecimal subtraction
 - Hexadecimal multiplication
 - Hexadecimal carry and borrow rules
-

3. Complements and Signed Number Representation

3.1 Complements

- r 's complement
- $(r - 1)$'s complement
- 1's complement
- 2's complement
- 9's complement
- 10's complement
- Shortcut method for 2's complement

3.2 Signed Number Representation

- Signed magnitude representation
- 1's complement representation
- 2's complement representation
- Biased representation
- Excess representation
- Sign bit
- Range of signed numbers
- Range of unsigned numbers

3.3 Arithmetic Using Complements

- Binary subtraction using 1's complement

- Binary subtraction using 2's complement
- Decimal subtraction using 9's complement
- Decimal subtraction using 10's complement
- Overflow detection in signed arithmetic
- Carry discard rule
- End-around carry

3.4 Important Hidden Topics

- Sign extension
 - Zero extension
 - Arithmetic overflow vs carry
 - Saturation arithmetic
 - Wrap-around arithmetic
 - Fixed-point representation
 - Floating-point basics
 - IEEE-754 introduction
 - Two's complement hardware advantage
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4. Digital Codes

4.1 Classification of Codes

- Weighted codes
- Non-weighted codes
- Sequential codes
- Cyclic codes
- Reflective codes
- Alphanumeric codes
- Error-detecting codes
- Error-correcting codes

4.2 Binary Coded Decimal Codes

- BCD code
- 8421 BCD code
- BCD representation
- BCD addition
- BCD correction using 0110
- Invalid BCD states
- BCD subtraction
- BCD to binary conversion
- Binary to BCD conversion
- Shift-add-3 method / double dabble algorithm

4.3 Weighted Codes

- 8421 code

- 2421 code
- 5211 code
- 5421 code
- Excess-3 code
- Excess-3 addition
- Self-complementing codes

4.4 Gray Code

- Introduction to Gray code
- Binary to Gray conversion
- Gray to binary conversion
- Reflected Gray code
- Cyclic property
- Applications of Gray code
- Gray code in rotary encoders
- Gray code in asynchronous FIFO pointers

4.5 Error Detection Codes

- Parity bit
- Even parity
- Odd parity
- Parity generator
- Parity checker
- Longitudinal redundancy check
- Checksum
- CRC basics

4.6 Error Correction Codes

- Hamming code
- Hamming distance
- Single-bit error correction
- Double-bit error detection
- Syndrome calculation
- SECDED code
- Error location and correction

4.7 Alphanumeric Codes

- ASCII code
 - Extended ASCII
 - Unicode basics
 - EBCDIC
 - Seven-segment display code
-

5. Boolean Algebra and Switching Theory

5.1 Introduction to Boolean Algebra

- Boolean variable
- Boolean constant
- Boolean function
- Literal
- Complement
- Boolean expression
- Logic operation

5.2 Basic Boolean Operations

- AND operation
- OR operation
- NOT operation
- NAND operation
- NOR operation
- XOR operation
- XNOR operation

5.3 Boolean Laws

- Identity law
- Null law
- Idempotent law
- Complement law
- Involution law
- Commutative law
- Associative law
- Distributive law
- Absorption law
- Consensus theorem
- Redundancy theorem
- De Morgan's theorems

5.4 Boolean Algebra Manipulation

- Simplification using laws
- Algebraic minimization
- Boolean expression expansion
- Boolean expression factorization
- Removing redundant terms
- Statement problems in Boolean algebra

5.5 SOP and POS Forms

- Sum of Products form

- Product of Sums form
- Standard SOP
- Standard POS
- Canonical SOP
- Canonical POS
- Minimal SOP
- Minimal POS
- Minterms
- Maxterms
- Conversion from truth table to SOP
- Conversion from truth table to POS
- Minimal to canonical conversion
- Canonical to minimal conversion

5.6 Advanced Boolean Concepts

- Duality theorem
 - Dual form
 - Self-dual functions
 - Complement of Boolean functions
 - Venn diagram representation
 - Switching circuits
 - Positive logic
 - Negative logic
 - Functional completeness
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6. Logic Gates

6.1 Basic Logic Gates

- AND gate
- OR gate
- NOT gate
- Buffer gate

6.2 Universal Logic Gates

- NAND gate
- NOR gate
- NAND as universal gate
- NOR as universal gate

6.3 Derived / Special Logic Gates

- XOR gate
- XNOR gate
- Tri-state buffer
- Open-drain gate

- Open-collector gate
- Schmitt trigger input gate
- Transmission gate
- Pass transistor logic gate

6.4 Gate-Level Representations

- Logic symbol
- Truth table
- Boolean equation
- Timing diagram
- CMOS implementation
- NAND-only implementation
- NOR-only implementation

6.5 Important Gate Parameters

- Fan-in
- Fan-out
- Noise margin
- Propagation delay
- Rise time
- Fall time
- Power dissipation
- Power-delay product
- Loading effect
- Logic threshold voltage

6.6 Logic Families

- RTL logic
 - DTL logic
 - TTL logic
 - CMOS logic
 - NMOS logic
 - PMOS logic
 - ECL logic
 - BiCMOS logic
 - Comparison of logic families
 - Voltage compatibility
 - Current sourcing and sinking
-

7. Boolean Function Minimization

7.1 Karnaugh Map Basics

- Introduction to K-map
- Need for minimization

- Adjacency principle
- Gray code ordering
- Cell numbering
- Minterm placement
- Maxterm placement

7.2 Two-Variable K-map

- 2-variable SOP simplification
- 2-variable POS simplification

7.3 Three-Variable K-map

- 3-variable SOP simplification
- 3-variable POS simplification
- Pair formation
- Quad formation
- Octet formation

7.4 Four-Variable K-map

- 4-variable K-map layout
- Grouping rules
- Edge wrapping
- Corner grouping
- Overlapping groups
- Essential prime implicants
- Non-essential prime implicants

7.5 Five-Variable and Six-Variable K-map

- 5-variable K-map
- 6-variable K-map
- Pair of K-maps method
- Practical limitations

7.6 Don't Care Conditions

- Don't care minterms
- Don't care maxterms
- Using don't cares for SOP
- Using don't cares for POS
- Invalid states as don't cares
- Excess code don't cares

7.7 Implicants and Prime Implicants

- Implicant
- Prime implicant
- Essential prime implicant

- Covering table
- Minimal cover

7.8 Tabular Minimization

- Quine-McCluskey method
- Prime implicant chart
- Petrick's method
- Comparison with K-map
- Computer-aided minimization

7.9 Hidden Practical Topics

- Static hazards
 - Dynamic hazards
 - Functional hazards
 - Hazard-free logic design
 - Glitches in combinational logic
 - Redundant logic for hazard removal
 - Timing-based errors in logic circuits
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8. Combinational Logic Circuits

8.1 Introduction to Combinational Circuits

- Definition
- Features
- Difference between combinational and sequential circuits
- Block diagram
- Design procedure
- Analysis procedure

8.2 Standard Design Procedure

- Problem statement
- Identify input and output variables
- Construct truth table
- Derive Boolean expression
- Simplify using K-map
- Implement using gates
- Verify using truth table
- Draw timing diagram

8.3 Common Supporting Topics

- Propagation delay
- Critical path
- Gate delay

- Logic depth
 - Glitch behaviour
 - Fan-out loading
 - Active-low implementation
 - NAND-only realization
 - NOR-only realization
 - MUX-based realization
 - Decoder-based realization
-

9. Arithmetic Circuits

9.1 Adders

- Half adder
- Full adder
- Full adder using half adders
- Full adder using NAND gates
- Full adder using NOR gates
- Full adder using MUX
- Full adder using decoder
- 4-bit parallel adder
- Ripple carry adder
- Carry propagation delay
- Carry lookahead adder
- Carry lookahead generator
- Carry select adder
- Carry skip adder
- Carry save adder
- Serial adder
- Parallel adder
- Prefix adders:
- Kogge-Stone adder
- Brent-Kung adder
- Sklansky adder
- Han-Carlson adder

9.2 Subtractors

- Half subtractor
- Full subtractor
- Full subtractor using half subtractors
- Full subtractor using NAND gates
- Full subtractor using NOR gates
- Full subtractor using DEMUX
- Binary subtraction using adder
- 2's complement subtractor

9.3 Adder-Subtractor Circuits

- Controlled adder-subtractor
- 1's complement adder-subtractor
- 2's complement adder-subtractor
- Overflow detection circuit
- Sign detection circuit
- Zero detection circuit
- Carry and borrow logic

9.4 Multipliers

- 2-bit multiplier using half adders
- Array multiplier
- Combinational multiplier
- Sequential multiplier
- Shift-and-add multiplier
- Booth multiplier
- Modified Booth multiplier
- Wallace tree multiplier
- Dadda multiplier
- Signed multiplier
- Unsigned multiplier

9.5 Dividers

- Binary divider
- Restoring divider
- Non-restoring divider
- Sequential divider
- Signed division
- Divide-by-zero detection

9.6 BCD Arithmetic Circuits

- BCD adder
- BCD subtractor
- BCD correction logic
- Excess-3 adder
- Decimal arithmetic unit

9.7 ALU Circuits

- 1-bit ALU
- 4-bit ALU
- n-bit ALU
- Arithmetic operations
- Logic operations
- Shift operations
- Compare operations

- Status flags:
 - Carry flag
 - Zero flag
 - Sign flag
 - Overflow flag
 - Parity flag
 - ALU control lines
 - ALU slice architecture
 - Bit-slice processor concept
-

10. Data Selection and Routing Circuits

10.1 Multiplexers

- Introduction to multiplexer
- 2:1 MUX
- 4:1 MUX
- 8:1 MUX
- 16:1 MUX
- 32:1 MUX
- n:1 MUX
- MUX truth table
- MUX Boolean equation
- MUX using gates
- MUX tree
- 4:1 MUX using 2:1 MUX
- 8:1 MUX using 2:1 MUX
- 8:1 MUX using 4:1 MUX
- 32:1 MUX using 8:1 MUX
- Boolean function implementation using MUX
- Full adder using MUX
- Logical expression from MUX
- MUX as universal logic element

10.2 Demultiplexers

- Introduction to DEMUX
- 1:2 DEMUX
- 1:4 DEMUX
- 1:8 DEMUX
- 1:16 DEMUX
- DEMUX truth table
- DEMUX Boolean equations
- DEMUX using gates
- Full subtractor using DEMUX
- DEMUX as decoder
- Data distribution circuits

10.3 Encoders

- Introduction to encoder
- 2^n -to-n encoder
- Decimal to BCD encoder
- Octal to binary encoder
- Hexadecimal to binary encoder
- Priority encoder
- 4:2 priority encoder
- 8:3 priority encoder
- Valid output indicator
- Encoder limitations
- Encoder truth table

10.4 Decoders

- Introduction to decoder
- n-to- 2^n decoder
- 2:4 decoder
- 3:8 decoder
- 4:16 decoder
- BCD to decimal decoder
- BCD to seven-segment decoder
- Decoder with enable input
- Full adder using decoder
- Boolean function implementation using decoder
- Decoder as DEMUX
- Memory address decoding

10.5 Code Converters

- Binary to Gray converter
- Gray to binary converter
- BCD to Excess-3 converter
- Excess-3 to BCD converter
- Binary to BCD converter
- BCD to binary converter
- Seven-segment display decoder
- Seven-segment display driver
- Common anode display
- Common cathode display

10.6 Comparators

- 1-bit comparator
- 2-bit comparator
- 4-bit comparator
- n-bit comparator
- Magnitude comparator
- Equality comparator

- Cascaded comparator
- Comparator using subtractor
- Comparator applications

10.7 Parity Circuits

- Even parity generator
 - Odd parity generator
 - Even parity checker
 - Odd parity checker
 - 4-bit parity generator
 - Error detection using parity
 - XOR tree implementation
-

11. Sequential Logic Circuits

11.1 Introduction to Sequential Circuits

- Definition
- Need for memory
- Feedback in digital circuits
- Present state
- Next state
- Output dependency
- Difference between combinational and sequential circuits

11.2 Types of Sequential Circuits

- Synchronous sequential circuits
- Asynchronous sequential circuits
- Clocked sequential circuits
- Unclocked sequential circuits
- Pulse-mode circuits
- Level-mode circuits

11.3 Clock Concepts

- What is a clock?
- Clock period
- Clock frequency
- Duty cycle
- Clock edge
- Rising edge
- Falling edge
- Clock skew
- Clock jitter
- Clock uncertainty
- Clock gating

- Clock enable

11.4 Latches

- SR latch using NOR gates
- SR latch using NAND gates
- Active-high SR latch
- Active-low SR latch
- Gated SR latch
- Gated D latch
- Transparent latch
- Latch enable
- Race condition in latches
- Latch timing diagram

11.5 Flip-Flops

- Difference between latch and flip-flop
- Edge-triggered flip-flop
- Level-triggered storage element
- Pulse-triggered flip-flop
- Master-slave flip-flop
- Positive-edge triggered flip-flop
- Negative-edge triggered flip-flop

11.6 SR Flip-Flop

- Introduction to SR flip-flop
- Logic symbol
- Truth table
- Characteristic table
- Excitation table
- Characteristic equation
- Timing diagram
- Invalid condition
- Applications

11.7 D Flip-Flop

- Introduction to D flip-flop
- Logic symbol
- Truth table
- Characteristic table
- Excitation table
- Characteristic equation
- Timing diagram
- D latch vs D flip-flop
- Behaviour of master-slave D flip-flop
- Applications

11.8 JK Flip-Flop

- Introduction to JK flip-flop
- Logic symbol
- Truth table
- Characteristic table
- Excitation table
- Characteristic equation
- Timing diagram
- Race-around condition
- Master-slave JK flip-flop
- Edge-triggered JK flip-flop
- Applications

11.9 T Flip-Flop

- Introduction to T flip-flop
- Logic symbol
- Truth table
- Characteristic table
- Excitation table
- Characteristic equation
- Timing diagram
- Frequency division using T flip-flop
- Applications

11.10 Flip-Flop Control Inputs

- Preset input
- Clear input
- Reset input
- Set input
- Synchronous reset
- Asynchronous reset
- Active-high reset
- Active-low reset
- Reset recovery time
- Reset removal time

11.11 Flip-Flop Conversion

- Need for flip-flop conversion
- SR to JK conversion
- SR to D conversion
- SR to T conversion
- JK to SR conversion
- JK to D conversion
- JK to T conversion
- D to SR conversion
- D to JK conversion

- D to T conversion
- T to SR conversion
- T to JK conversion
- T to D conversion
- Five-step conversion method
- Conversion using excitation tables

11.12 Hidden Sequential Topics

- Setup time
 - Hold time
 - Clock-to-Q delay
 - Metastability
 - Synchronizers
 - Two-flop synchronizer
 - Reset synchronizer
 - Clock domain crossing basics
 - Pulse synchronization
 - Data synchronization
 - Enable vs clock gating
 - Race vs hazard vs metastability
-

12. Finite State Machines

12.1 FSM Fundamentals

- State
- Input
- Output
- Transition
- Present state
- Next state
- Initial state
- Final state
- State variable
- State encoding

12.2 State Representations

- State table
- State diagram
- State equation
- Transition table
- Output table
- Timing diagram

12.3 Mealy and Moore Machines

- Mealy machine
- Moore machine
- Comparison between Mealy and Moore machines
- Timing of outputs
- Glitches in Mealy outputs
- State reduction possibility

12.4 FSM Design Procedure

- Problem statement
- Identify inputs and outputs
- Define states
- Draw state diagram
- Create state table
- Choose flip-flop type
- Derive excitation table
- Derive next-state logic
- Derive output logic
- Minimize using K-map
- Implement circuit
- Verify using timing diagram

12.5 FSM Analysis

- Analysis using D flip-flop
- Analysis using JK flip-flop
- Analysis using T flip-flop
- Deriving state equation
- Deriving state table
- Deriving state diagram
- Identifying unused states
- Lockout condition

12.6 State Reduction and Assignment

- Equivalent states
- State reduction
- Partition method
- State assignment
- Binary encoding
- Gray encoding
- One-hot encoding
- Output-based assignment
- Unused state handling

12.7 Sequence Detectors

- Sequence detector basics

- Overlapping sequence detector
- Non-overlapping sequence detector
- Mealy sequence detector
- Moore sequence detector
- Example: 1001 detector
- Example: 1010 detector
- Example: 1101 detector
- State minimization for sequence detectors

12.8 Algorithmic State Machines

- ASM chart
- State box
- Decision box
- Conditional output box
- ASM chart for Moore machine
- ASM chart for Mealy machine
- ASM to hardware conversion
- Control unit design using ASM

12.9 Hidden FSM Topics

- Safe FSM design
 - Unreachable states
 - Deadlock states
 - FSM lockup
 - Default state recovery
 - Glitch-free FSM outputs
 - Registered outputs
 - FSM reset strategy
 - Formal verification of FSMs
-

13. Counters

13.1 Counter Fundamentals

- Introduction to counters
- Modulus of counter
- Count sequence
- Up counting
- Down counting
- Up/down counting
- Terminal count
- Counter states
- State diagram of a counter

13.2 Asynchronous / Ripple Counters

- Ripple counter basics
- 2-bit asynchronous up counter
- 3-bit asynchronous up counter
- 4-bit asynchronous up counter
- Asynchronous down counter
- 3-bit asynchronous down counter
- 4-bit asynchronous down counter
- Asynchronous up/down counter
- Propagation delay in ripple counters
- Glitches in ripple counters
- Decoding errors in ripple counters

13.3 Synchronous Counters

- Synchronous counter basics
- 2-bit synchronous up counter
- 3-bit synchronous up counter
- 4-bit synchronous up counter
- Synchronous down counter
- Synchronous up/down counter
- Design using D flip-flop
- Design using JK flip-flop
- Design using T flip-flop
- Excitation table method
- K-map simplification for counter design

13.4 Mod Counters

- MOD-n counter
- MOD-2 counter
- MOD-4 counter
- MOD-5 counter
- MOD-6 counter
- MOD-10 counter
- Decade counter
- BCD ripple counter
- BCD synchronous counter
- Counter reset logic
- Counting up to a particular value

13.5 Special Counters

- Ring counter
- Johnson counter
- Twisted ring counter
- One-hot counter
- Gray code counter
- Binary counter

- LFSR counter
- Pseudo-random sequence counter

13.6 Counter Applications

- Frequency divider
- Digital clock
- Event counter
- Timer
- Stopwatch
- Address counter
- Program counter
- Sequence generator
- PWM generation

13.7 Hidden Counter Topics

- Counter lockout
 - Self-starting counter
 - Unused state recovery
 - Glitch-free decoding
 - Cascading counters
 - Synchronous clear vs asynchronous clear
 - Terminal count detection
 - Clock enable counter design
-

14. Registers and Shift Registers

14.1 Register Basics

- Introduction to registers
- Register definition
- n-bit register
- Parallel load register
- Register with enable
- Register with reset
- Data storage using flip-flops

14.2 Data Formats

- Serial data
- Parallel data
- Serial input
- Parallel input
- Serial output
- Parallel output
- Data word
- Register classification

14.3 Shift Registers

- SISO shift register
- SIPO shift register
- PISO shift register
- PIPO register
- Left shift register
- Right shift register
- Bidirectional shift register
- Universal shift register

14.4 Special Register Circuits

- Ring register
- Shift counter
- Barrel shifter
- Rotate left circuit
- Rotate right circuit
- Arithmetic shift circuit
- Logical shift circuit

14.5 Register Applications

- Serial-to-parallel conversion
- Parallel-to-serial conversion
- Delay line
- Data buffer
- Temporary storage
- Communication interface
- Sequence generator
- Arithmetic shift operations

14.6 Hidden Register Topics

- Register transfer operation
 - Register transfer notation
 - Load enable
 - Clock enable
 - Reset behaviour
 - Shift enable
 - Hold condition
 - Multi-register datapath
 - Bus-connected registers
-

15. Memory Circuits

15.1 Memory Fundamentals

- Memory cell
- Word
- Address
- Data input
- Data output
- Read operation
- Write operation
- Memory capacity
- Memory organization
- Memory map

15.2 ROM

- ROM
- PROM
- EPROM
- EEPROM
- Flash memory
- Mask ROM
- ROM truth table
- ROM as combinational logic
- Lookup table implementation

15.3 RAM

- RAM
- SRAM
- DRAM
- Static memory cell
- Dynamic memory cell
- Read cycle
- Write cycle
- Refresh operation
- Memory timing

15.4 Memory Decoding

- Address decoder
- Row decoder
- Column decoder
- Chip select
- Output enable
- Write enable
- Memory expansion
- Word expansion

- Bit expansion

15.5 Special Memories

- FIFO
- LIFO / stack
- Register file
- Dual-port RAM
- Content-addressable memory
- Cache memory basics

15.6 Hidden Memory Topics

- Memory timing diagram
 - Read access time
 - Write recovery time
 - Setup and hold for memory
 - Bus contention
 - Tri-state data bus
 - Byte enable signals
 - Endianness
 - Memory-mapped I/O
-

16. Programmable Logic Devices

16.1 Introduction to Programmable Logic

- Need for programmable logic
- Fixed logic vs programmable logic
- Programmable interconnect
- Programmable logic arrays

16.2 ROM-Based Logic

- ROM as logic implementation
- PROM
- EPROM
- EEPROM
- LUT concept

16.3 PLA

- Programmable Logic Array
- Programmable AND plane
- Programmable OR plane
- PLA structure
- PLA implementation of SOP
- PLA advantages and disadvantages

16.4 PAL

- Programmable Array Logic
- Programmable AND plane
- Fixed OR plane
- PAL structure
- PAL vs PLA
- PAL limitations

16.5 CPLD

- CPLD architecture
- Macrocell
- Product term logic
- Interconnect matrix
- CPLD applications

16.6 FPGA

- FPGA basics
- LUT
- Flip-flop
- Configurable logic block
- Switch matrix
- I/O block
- Block RAM
- DSP block
- Clock management block
- FPGA configuration memory
- FPGA design flow

16.7 Hidden PLD Topics

- LUT truth table mapping
- Technology mapping
- Logic synthesis
- Place and route
- Timing closure
- Bitstream generation
- FPGA vs ASIC comparison

17. Data Path and Control Path Circuits

17.1 Datapath Basics

- Register
- MUX
- ALU

- Bus
- Shifter
- Comparator
- Accumulator
- Register file

17.2 Control Path Basics

- Control signals
- Control word
- FSM controller
- Hardwired control
- Microprogrammed control
- Instruction decoding

17.3 Register Transfer Level

- RTL concept
- Register transfer operation
- Micro-operations
- Arithmetic micro-operations
- Logic micro-operations
- Shift micro-operations
- Bus transfer

17.4 Common Datapath Blocks

- Accumulator
- Program counter
- Instruction register
- Memory address register
- Memory data register
- Status register
- General-purpose register file

17.5 Hidden Datapath Topics

- Control signal timing
 - Data hazard
 - Resource sharing
 - MUX-based datapath design
 - Bus contention
 - Tri-state bus vs MUX bus
 - Pipelined datapath basics
-

18. Digital Timing and Practical Circuit Issues

18.1 Delay Concepts

- Propagation delay
- Contamination delay
- Rise time
- Fall time
- Transition time
- Critical path
- Maximum frequency
- Path delay

18.2 Timing Constraints

- Setup time
- Hold time
- Clock-to-Q delay
- Clock skew
- Clock jitter
- Slack
- Timing violation
- Recovery time
- Removal time

18.3 Glitches and Hazards

- Static-0 hazard
- Static-1 hazard
- Dynamic hazard
- Functional hazard
- Hazard detection using K-map
- Hazard removal
- Glitch-free design

18.4 Clocking Issues

- Clock distribution
- Clock skew
- Clock gating
- Clock enable
- Clock domain crossing
- Synchronizer
- Metastability
- Reset synchronization

18.5 Signal Integrity Basics

- Fan-in

- Fan-out
 - Loading
 - Noise margin
 - Crosstalk
 - Ground bounce
 - Power supply noise
 - Floating inputs
 - Pull-up and pull-down resistors
-

19. Input/Output Interface Circuits

19.1 Switch Interface

- Mechanical switch
- Pull-up resistor
- Pull-down resistor
- Switch bounce
- Hardware debounce
- Software debounce
- Schmitt trigger input

19.2 LED Interface

- LED driving
- Current-limiting resistor
- Active-high LED
- Active-low LED
- Seven-segment display driving

19.3 Display Circuits

- Seven-segment display decoder
- Common anode display
- Common cathode display
- Multiplexed seven-segment display
- Display refresh
- BCD to seven-segment decoder

19.4 Bus Interface

- Tri-state buffer
- Bus driver
- Bus receiver
- Bidirectional bus
- Bus contention
- Open-drain bus
- Wired-AND logic
- I²C-style open-drain concept

19.5 Digital Communication Basics

- Serial communication
 - Parallel communication
 - UART basics
 - SPI basics
 - I²C basics
 - Handshaking signals
 - Ready-valid interface
 - Strobe signal
 - Acknowledge signal
-

20. Digital System Building Blocks

20.1 Simple Digital Systems

- Digital clock
- Stopwatch
- Frequency counter
- Event counter
- Sequence generator
- Traffic light controller
- Elevator controller
- Vending machine controller
- Password lock system
- Electronic voting machine
- Digital dice
- Reaction timer

20.2 Processor-Oriented Blocks

- ALU
- Register file
- Program counter
- Instruction register
- Control unit
- Data memory
- Instruction memory
- Decoder
- Immediate generator
- Branch comparator
- Load-store unit

20.3 Communication-Oriented Blocks

- UART transmitter
- UART receiver
- SPI master

- SPI slave
- I²C controller
- FIFO buffer
- Serializer
- Deserializer

20.4 DSP-Oriented Digital Blocks

- Adder tree
 - Multiplier
 - MAC unit
 - Accumulator
 - FIR filter
 - IIR filter basics
 - Digital integrator
 - Digital differentiator
 - Numerically controlled oscillator
-

21. HDL-Based Digital Design

21.1 HDL Introduction

- Need for HDL
- Verilog basics
- SystemVerilog basics
- VHDL basics
- Simulation vs synthesis

21.2 Verilog Modelling Styles

- Gate-level modelling
- Dataflow modelling
- Behavioural modelling
- Structural modelling
- RTL modelling

21.3 Combinational HDL Design

- Continuous assignment
- always_comb block
- Blocking assignment
- Avoiding unintended latches
- Case statement
- If-else statement
- Full case coverage

21.4 Sequential HDL Design

- always_ff block
- Non-blocking assignment
- Synchronous reset
- Asynchronous reset
- Clock enable
- Register inference
- Flip-flop inference

21.5 HDL Design Examples

- Logic gates
- Half adder
- Full adder
- Multiplexer
- Decoder
- Comparator
- Flip-flops
- Counter
- Shift register
- FSM
- ALU
- FIFO

21.6 Hidden HDL Topics

- Blocking vs non-blocking assignment
 - Race conditions in simulation
 - Sensitivity list errors
 - Latch inference
 - Multiple drivers
 - X propagation
 - Z state
 - Parameterized design
 - Generate blocks
 - Testbench basics
-

22. Digital Verification and Testing

22.1 Basic Verification

- Testbench
- Stimulus
- Expected output
- Monitor
- Checker
- Waveform analysis

- Self-checking testbench

22.2 Functional Verification

- Directed testing
- Random testing
- Constrained random testing
- Functional coverage
- Code coverage
- Assertions
- Scoreboard

22.3 Timing Verification

- Static timing analysis
- Setup check
- Hold check
- Maximum frequency calculation
- Critical path analysis
- Timing simulation

22.4 Hardware Testing

- Stuck-at fault
- Bridging fault
- Transition fault
- Scan chain
- Boundary scan
- JTAG basics
- Built-in self-test
- Design for testability

22.5 Hidden Testing Topics

- Test vectors
- Fault coverage
- Controllability
- Observability
- ATPG
- Signature analysis
- Logic analyzer usage
- Oscilloscope timing capture

23. CMOS Digital Circuit Basics

23.1 MOSFET as a Switch

- NMOS switch

- PMOS switch
- CMOS inverter
- Pull-up network
- Pull-down network
- Complementary logic

23.2 CMOS Logic Gates

- CMOS NAND
- CMOS NOR
- CMOS AND
- CMOS OR
- CMOS XOR
- Transmission gate logic
- Pass transistor logic

23.3 CMOS Characteristics

- Switching threshold
- Noise margin
- Propagation delay
- Dynamic power
- Static power
- Short-circuit power
- Leakage power
- Fan-out

23.4 Hidden CMOS Topics

- Body effect
- Charge sharing
- Ratioed logic
- Dynamic logic
- Domino logic
- Clocked CMOS
- Power gating
- Clock gating
- Level shifters

24. Advanced and Complex Digital Circuits

24.1 Arithmetic Processing Circuits

- Advanced ALU
- Multiply-accumulate unit
- Floating-point adder
- Floating-point multiplier
- Saturating arithmetic unit

- SIMD datapath basics

24.2 Memory and Buffer Circuits

- FIFO
- Asynchronous FIFO
- Circular buffer
- Register file
- Cache controller basics
- Memory controller basics

24.3 Processor Circuits

- Simple CPU datapath
- Hardwired control unit
- Microprogrammed control unit
- Instruction decoder
- Pipeline registers
- Branch unit
- Hazard detection unit
- Forwarding unit

24.4 Digital Communication Circuits

- UART
- SPI
- I²C
- Serializer/deserializer
- CRC generator
- Packet parser
- Line encoder
- Manchester encoder
- 8b/10b encoder basics

24.5 Digital Control Circuits

- PWM generator
- PID controller basics
- Motor control logic
- Traffic light controller
- Vending machine controller
- Elevator controller
- Washing machine controller

24.6 Cryptographic and Security Circuits

- LFSR
- Pseudo-random number generator
- Stream cipher basics
- AES round structure basics

- Hash function hardware basics
 - Secure key register
 - True random number generator basics
-

25. Practice and Problem-Solving Sections

25.1 Boolean Algebra Practice

- Simplification problems
- SOP/POS conversion
- Canonical form problems
- Dual and complement problems
- Statement-to-expression problems

25.2 K-map Practice

- 2-variable K-map
- 3-variable K-map
- 4-variable K-map
- 5-variable K-map
- Don't care problems
- Maxterm K-map problems
- Hazard-free K-map problems

25.3 Combinational Circuit Practice

- Adder design
- Subtractor design
- MUX implementation
- DEMUX implementation
- Encoder design
- Decoder design
- Comparator design
- Code converter design
- Seven-segment display problems

25.4 Sequential Circuit Practice

- Flip-flop truth table problems
- Characteristic table problems
- Excitation table problems
- Flip-flop conversion problems
- Counter design
- Register design
- Sequence detector design
- FSM design

25.5 Timing Practice

- Propagation delay calculation
- Maximum frequency calculation
- Setup time check
- Hold time check
- Clock skew problems
- Counter delay problems
- Critical path problems

25.6 HDL Practice

- Verilog code for gates
 - Verilog code for adders
 - Verilog code for MUX/DEMUX
 - Verilog code for encoder/decoder
 - Verilog code for flip-flops
 - Verilog code for counters
 - Verilog code for shift registers
 - Verilog code for FSMs
 - Testbench writing
-

26. Quick Master Checklist of Digital Circuits

26.1 Basic Gate-Level Circuits

- Buffer
- NOT
- AND
- OR
- NAND
- NOR
- XOR
- XNOR
- Tri-state buffer
- Schmitt trigger
- Transmission gate

26.2 Combinational Circuits

- Half adder
- Full adder
- Ripple carry adder
- Carry lookahead adder
- Carry select adder
- Half subtractor
- Full subtractor
- Adder-subtractor

- Multiplier
- Divider
- Comparator
- Parity generator
- Parity checker
- MUX
- DEMUX
- Encoder
- Priority encoder
- Decoder
- Code converter
- Seven-segment decoder
- BCD adder
- ALU

26.3 Sequential Circuits

- SR latch
- D latch
- SR flip-flop
- D flip-flop
- JK flip-flop
- T flip-flop
- Master-slave flip-flop
- Edge-triggered flip-flop
- Register
- Shift register
- Counter
- Ring counter
- Johnson counter
- Sequence detector
- FSM controller

26.4 Memory Circuits

- ROM
- PROM
- EPROM
- EEPROM
- Flash
- SRAM
- DRAM
- FIFO
- Register file
- Cache memory

26.5 Programmable Logic Circuits

- ROM logic
- PLA
- PAL

- CPLD
- FPGA
- LUT
- CLB
- Macrocell

26.6 System-Level Digital Circuits

- ALU
- Datapath
- Control unit
- CPU basics
- UART
- SPI
- I²C
- PWM generator
- Timer
- Digital clock
- Traffic light controller
- Vending machine controller
- FIR filter
- MAC unit

27. Important “Hidden” Topics That Should Not Be Missed

- Active-high and active-low logic
- Positive and negative logic
- Floating inputs
- Pull-up and pull-down resistors
- Debouncing
- Fan-in and fan-out
- Noise margin
- Propagation delay
- Contamination delay
- Setup time
- Hold time
- Clock skew
- Clock jitter
- Metastability
- Synchronizers
- Clock domain crossing
- Reset domain crossing
- Glitches
- Hazards
- Race condition
- Race-around condition
- Bus contention

- Tri-state logic
 - Open-drain logic
 - Wired-AND logic
 - Unused states
 - Lockout condition
 - Self-starting counters
 - Safe FSM design
 - State encoding
 - One-hot encoding
 - Gray encoding
 - Synchronous reset
 - Asynchronous reset
 - Clock gating
 - Clock enable
 - Critical path
 - Maximum clock frequency
 - Timing closure
 - HDL simulation vs synthesis mismatch
 - Blocking vs non-blocking assignments
 - Latch inference
 - Multiple drivers
 - X and Z logic states
 - Testbench design
 - Assertions
 - Static timing analysis
 - Design for testability
 - Scan chain
 - JTAG
 - FPGA LUT mapping
 - ASIC standard cells
 - Power consumption in digital circuits
 - Dynamic power
 - Static power
 - Leakage power
 - Power-delay product
 - CMOS implementation of gates
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